Philipp Michel

SVP of Autonomy & AI, Motional

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PROFILE

Technology executive leading machine learning at scale in the physical world. Leads the science, engineering, and strategic direction behind Motional's autonomous driving technology. Previously held senior AI and autonomy leadership roles at Cruise / GM, Amazon, Toyota, and Google / X. Scaled Amazon's autonomous delivery to tens of thousands of packages and built Toyota Research Institute's machine learning team from scratch. Former Vice President and Head of Asia Quantitative Volatility Trading at Goldman Sachs in Tokyo. Ph.D. / M.S. in Robotics, Carnegie Mellon; M.A. / B.A. in Computer Science, Cambridge.

Expertise: AI & ML · Data & Infrastructure · Autonomy · Robotics · Product & Strategy · Derivatives Trading

EXPERIENCE

Motional Seattle, WA — 2026 – present

SVP of Autonomy & AI

- Lead the science, engineering, and strategic direction behind Motional's autonomous driving technology, across ML, autonomy, data, and infrastructure

Cruise / General Motors Seattle, WA — 2023 – 2026

Distinguished Engineer (General Motors) — 2025 – 2026

- Drove technical strategy across Embodied AI

Principal AI / ML Engineer · Principal Applied Scientist — 2024 – 2025

- Drove technical strategy across Simulation, Data, and Testing

Technical Advisor to the CTO / SVP of AI — 2023 – 2024

- Shaped AI / ML technical strategy, identified key investment areas, drove execution

Amazon Seattle, WA — 2019 – 2023

Head of Perception & Planning — 2020 – 2023

- Led autonomy science / engineering teams across perception and planning
- Scaled autonomous delivery to tens of thousands of packages across four US states

Principal Product Manager – Technical — 2019 – 2020

- Amazon Scout: Amazon's robotic last-mile delivery solution
- Managed autonomy technology roadmap & execution
- Led autonomy product / program management teams

Toyota Motor Corporation Worldwide — 2017 – 2019

Global AI Strategy Advisor

- Strategic / technical advisor to TMC's global artificial intelligence initiatives
- Coordinated investment and research / commercial partnerships
- Technical leadership for TMC's AI efforts in Japan

Toyota Research Institute Palo Alto, CA — 2016 – 2017

Senior Manager, Machine Intelligence

- Devised and executed TRI's machine learning strategy
- Built world-class ML team from scratch
- Led ML algorithms, infrastructure, model / data management and on-platform deployment efforts
- Focused on perception for autonomous vehicles and robotics
- Led business development, investment evaluation, and research partnerships

Google, Inc. / Alphabet, Inc. Mountain View, CA — 2015 – 2016

Senior Roboticist / Software Engineer

- Advanced perception / computer vision deployed on multiple confidential robotic platforms
- Object detection & real-time tracking (monocular and depth-based) for manipulation and navigation (wheeled and legged)
- High frame-rate localization / pose estimation

Goldman Sachs Japan, Co. Ltd. Tokyo, Japan 2008 – 2015

Vice President, Equity Derivatives Trading

- Head, Asia Quantitative Volatility Trading. Low latency, high Sharpe ratio algorithmic options trading in Japan, Korea, and India (2013 – 2014)
- Traded the firm's index derivatives book in Japan (Nikkei, TOPIX, etc.). Consistently high yearly P&L (2008 – 2013)
- Specialized in pricing / hedging / market making volatility and delta-one products including options, light exotics, var / vol swaps, div swaps, forwards, rolls, etc.
- Actively traded cross-country positions in S&P, Euro Stoxx and Asian underlying assets
- Trading lead for several automation projects in pricing, market making, systematic trading / hedging

Earlier Experience

UBS, Summer Associate, Equities Trading Stamford, CT / New York, NY 2007

Wolfram Research, Developer, Special Projects Office Waltham, MA 2004

Goldman Sachs International, Summer Analyst London, UK 2002

ACADEMIC & ADVISORY APPOINTMENTS

The University of Tokyo Tokyo, Japan 2018 – 2024

Project Assistant Professor

- Appointed to Next Generation Artificial Intelligence Research Center; jointly appointed to Department of Mechano-Informatics
- Artificial intelligence and robotics research and advisory

The Partnership on AI San Francisco, CA 2018 – 2019

Committee Member

- Representing The University of Tokyo
- Member of the committees on Safety-Critical AI and on Collaborations between People and AI Systems

AIST Digital Human Research Center Tokyo, Japan 2005 – 2008

Visiting Researcher, Humanoid Robotics

- Postdoctoral Fellow (2008) and Summer Fellow (2005), Japan Society for the Promotion of Science

EDUCATION

Carnegie Mellon University Pittsburgh, PA 2004 – 2008

Ph.D. & M.S., Robotics

- Robotics Institute, School of Computer Science · Summa cum laude (GPA 4.00)
- Research: Humanoid Robotics, Computer Vision · 15+ peer-reviewed publications

Yale University New Haven, CT 2003 – 2004

Postgraduate Fellow (Robotics)

University of Cambridge Cambridge, UK 2000 – 2003

M.A. & B.A. (Hons), Computer Science

- Triple First Class Honours (summa cum laude)

HONORS

- Nvidia Fellowship (2006 – 2007)
- Triple Scholar of Churchill College Cambridge for outstanding academic performance
- Fellow of Cambridge University European Trust
- Scholar of the German National Merit Foundation (top 0.2% of students)
- Cambridge University prize for outstanding undergraduate dissertation

ADDITIONAL

Languages German (native) · English (native-level) · Spanish (native-level) · Japanese (intermediate) · Greek (conversational)

Athletics CrossFit Level 1 Trainer · Competitive rowing & water polo at Cambridge University · Nationally competitive high school swimming in Costa Rica